Lasitha Amarasinghe

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Interests

Agentic AI, Machine Learning, Artificial Intelligence, Computer Vision, Deep Learning

Professional Experience

Software Engineer (AI/ML) - Kaya, USA Jun 2026 - Present

AI-Agentic Hiring Automation Platform

- Automated candidate sourcing, CV scoring, and shortlisting, integrated with Zoho Recruit.
- Engineered multi-agent CV-scoring workflows on the Kaya AI platform, with custom scoring logic, threshold-based filtering, and asynchronous request orchestration to process large candidate pools.

AI-Agentic Healthcare Provider Data Management Platform (USA client)

- Developed backend services automating roster ingestion, validation, and transformation.
- Built and extended agentic AI workflows for document classification, data validation, rule execution, and status reporting, reducing manual effort in provider data operations.
- Integrated external verification services and enterprise systems to agentic workflows.
- Collaborated with cross-functional teams in an Agile environment.

Tech Stack: AI Agents, Python, FastAPI, Next.js, React.js, TypeScript, Celery, PostgreSQL, Redis, Docker

Research Intern - National University of Singapore (NUS) Dec 2024 - May 2025

Supervisor: Prof. David Hsu - Director, Smart Systems Institute, NUS Intern Certificate

• Architected a context-aware multimodal Augmented Reality agent for the TOM project fusing Vision, Gaze, and Speech inputs to enable natural interaction on HoloLens 2.
• Engineered a high-performance data pipeline connecting Unity and Python via Protocol Buffers and WebSockets, optimizing throughput for real-time sensor streaming.
• Engineered a deterministic gaze-trigger to filter user intent, reducing API calls and minimizing latency.
Tech Stack - Python, LangChain, Unity, AI Agents, WebSockets, Protocol Buffers, Augmented Reality

Undergraduate Teaching Assistant - University of Moratuwa Sep 2025 - Dec 2025

• Mentored 100+ undergraduates during lab sessions for Internet of Things & Laboratory Practice modules.

Education

University of Moratuwa, Sri Lanka Mar 2022 - June 2026

B.Sc. Eng.(Hons.) in Electronic & Telecommunication Engineering | CGPA: 3.88/4.00 Transcript
Minor in Mathematics | First Class Honours | Dean's List: 7 out of 8 Semesters
Most Outstanding Graduate, Faculty of Engineering, University of Moratuwa - Nominee

Coursework: Pattern Recognition, Image Processing & Machine Vision, Deep Learning for Vision, Software Design Competition, Data Structures & Algorithms, Linear Algebra, Multivariable Calculus, Applied Statistics

Dharmaraja College, Kandy, Sri Lanka Jan 2012 - Dec 2020

G.C.E. A/L , AAA , Z-Score: 2.5972 , Island Rank: 100/35000 (Top 0.3%) Transcript

Publications

A Privacy-Preserving Multi-Stage Framework for Contactless Neonatal Jaundice Classification and Bilirubin Estimation*, Moratuwa Engineering Research Conference (MERCon 2026)*

NMVP: Privacy-Preserving Neonatal Vital Sign Estimation via Remote Photoplethysmography*, IEEE Journal of Biomedical & Health Informatics (JBHI) (under review)*

Technical Demonstrations

AiGet: Transforming Everyday Moments into Hidden Knowledge Discovery,
Annual Singapore HCI Meetup, Singapore Management University (SMU), 2025

Skills

Programming Languages	Python, Java, C, C++, JavaScript, SQL, MATLAB
Deep Learning	TensorFlow, PyTorch, Keras, ONNX, TensorRT
Generative AI	LangChain, LangGraph, Hugging Face, LlamaIndex, Qdrant
AI & Data	Pandas, NumPy, OpenCV, Scikit-learn, Matplotlib
Web & Backend	React.js, Node.js, Spring Boot, FastAPI, Redis, PostgreSQL
MLOps & Cloud	Git, Docker, MLflow, BitBucket, AWS, Linux

Research Projects

Contactless Neonatal Monitoring System (FYP) *(Python, PyTorch, Computer Vision)* Link

• Developed a contactless neonatal health monitoring system using Remote Photoplethysmography (rPPG) to estimate heart rate, SpO_2, and jaundice levels of neonates by physiological signal extraction.
• Preserved neonatal privacy by anonymizing facial features and on-device processing.
• Achieved SOTA performance in contactless neonatal HR estimation with a MAE of 2.79 bpm on NBHR dataset.
• Achieved an accuracy of 85% and a F1 Score of 80% for neonatal jaundice classification on NJN dataset.
• Optimized models for mobile inference with TFLite to ensure low latency performance and patient data privacy.
• Neonatal facial video dataset collection from 66 neonates at De Soysa Maternity Hospital, Colombo
• Performed data cleaning and signal reconstruction on neonatal facial videos to mitigate noise and artifacts.

Projects

CSE Smart Scout - Hierarchical Multi-Agent System *(LangGraph, Python, REST APIs)* Link

• Architected a Supervisor-Worker Multi-Agent System coordinating specialized quantitative and qualitative agents, using persistent state checkpointing to enable deterministic routing and fault-tolerant execution.
• Engineered a dual-layer safety node combining PII sanitization with financial regulatory guardrails, intercepting and masking sensitive entities while filtering non-compliant investment claims.
• Implemented dynamic Human-in-the-Loop (HITL) review via LangGraph interrupt patterns, halting downstream state transitions for human analyst verification on high-risk trade signals.

Hybrid Agentic RAG Assistant (Individual) *(LangChain, ChromaDB, Groq LPU, Docker)* Link

• Engineered a hybrid RAG architecture using LangChain and ChromaDB, decoupling local Hugging Face embeddings from Groq (Llama 3.1) inference to achieve faster generation speeds with zero API costs.
• Developed a containerized agentic router dynamically switching between web search and local context, ensuring consistent deployment environments and seamless scalability.

TeaLeaf Lens - Tea Quality Inspection System *(TensorFlow, FastAPI, TFLite, MLflow, AWS)* Link

• Developed a lightweight tea pathology detector using MobileNetV3, achieving 78% accuracy and 9x compression via TFLite quantization for deployment on edge devices and utilized MLflow for full experiment tracking.
• Deployed an end-to-end cloud inference system on AWS EC2 integrating a FastAPI backend and Streamlit dashboard, optimizing server stability.
• Engineered a robust data augmentation pipeline (brightness, rotation) to improve model generalization.

UAV Surveillance & Payload Identification System (Team) *(YOLOv8, Image Fusion)* Link

• Architected a real-time multimodal surveillance pipeline fusing dual-stream sensor data (RGB & Thermal), ensuring continuous system availability and tracking persistence even under degraded environmental conditions.
• Implemented a Y-shape Dynamic Transformer (YDTR) to fuse thermal and visual data streams, integrating YOLOv8 and BoT-SORT to maintain robust detection and tracking.

Transformer Inspection System (Team) *(Java Spring Boot, React.js, Python, PostgreSQL)* Link

• Developed a scalable maintenance platform, digitizing legacy workflows for high-voltage transformer inspection.
• Integrated Python-based Computer Vision models with the Spring Boot backend through REST APIs.
• Integrated a thermal defect detection pipeline directly into the business logic, reducing manual review time for thermal anomalies by automating hazard identification.

Achievements

- **IEEEXtreme 19.0:** *Sri Lanka Rank 4, World Rank 76, 8500+ teams, 24 hours* - IEEE
- **IEEEXtreme 18.0:** *Sri Lanka Rank 8, World Rank 129, 8500+ teams, 24 hours* - IEEE
- **Techno Xtreme 2023:** *Champions, 250+ teams* - IESL Student Chapter, University of Moratuwa
- **CYPHER 2023:** *Champions, 200+ teams* - IEEE WIE Student Branch, Kotelawala Defence University
- **Brainstorm 2026:** *Champions, 100+ teams* - IEEE EMBS Student Branch, University of Moratuwa
- **ICE 2025:** *Best Team Dynamics Award* - Augmented Human Lab, National University of Singapore
- **AlgoXplore:** *3rd place, 250+ teams* - Faculty of Computing, NSBM Green University
- **SLIoT Challenge:** *4th place, 100+ teams* - Computer Science & Engineering, University of Moratuwa
- **ENIGMA 2024:** *5th place, 250+ teams* - Mathematics Society, University of Moratuwa
- **Data Crunch 2025:** *5th place, 250+ teams* - Computer Science & Engineering, University of Moratuwa

Certifications

- **Deep Learning Specialization**, DeepLearning.AI & Stanford Online via Coursera *2024*
- **Machine Learning Specialization**, DeepLearning.AI & Stanford Online via Coursera *2023*
- **Calculus for Machine Learning and Data Science**, DeepLearning.AI & via Coursera *2023*
- **Building Transformer-Based NLP Applications**, NVIDIA Deep Learning Institute *2026*
- **TensorFlow-Keras-PyTorch Bootcamps**, OpenCV University *2025*

Technical Articles

- **Production-Ready Agents: Implementing PII Protection and HITL Guardrails** Link
- **The Agent Integration Nightmare: How MCP Changes Everything** Link

Leadership

Vice President - Spark Branch - Electronic Club of University of Moratuwa Link

• Spearheaded operations to drive technical innovation and professional development for undergraduates.
• Organized Spark Challenge, achieving Sri Lanka's record- highest prize pool for a university competition.

Department Representative - Electronic and Telecommunication Engineering Link

• Coordinated the planning and logistics for 10+ major events, including EXMO'23 Flagship Exhibition.
• Bridged communication between faculty and students to align academic initiatives.

Organizer - Smart Glasses Bootcamp - National University of Singapore Link

• Organized a 6-day bootcamp on wearable computing & augmented reality for over 40 international participants.
• Facilitated hands-on learning, interactive workshops, and technical demonstrations.

Referees

Dr. Sampath K. Perera
B.Sc.Eng.(Moratuwa), M.Sc.(Western), Ph.D.(RUB)
Senior Lecturer, ENTC, University of Moratuwa
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Mr. Koh Chun Keat
Head of Engineering, Smart Systems Institute,
National University of Singapore, Singapore
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